

- Q.19 Write the significance of Reynolds number and Prandtl number. (CO3)
- Q.20 Discuss about the Stefan-Boltzman law. (CO4)
- Q.21 Explain the concept of Plank's law. (CO4)
- Q.22 Explain about the Radiation shield. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Drive the expression with assumptions for Steady-state one dimensional heat conduction through a composite wall. (CO2)
- Q.24 What is dimensional Analysis? Explain and drive the application of dimensional analysis for natural convection $\{Nu = f(Gr, Pr)\}$ (CO3)
- Q.25 Write short note on the following: (CO4)
- Monochromatic emissivity of Black-body
 - Absorptivity.

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Roll No.

4th Sem / Chemical

Subject : Heat Transfer Operation - I

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 How many modes are in heat transfer? (CO1)
- One
 - Two
 - Three
 - Four
- Q.2 Which of the following is not a method of heat transfer? (CO1)
- Conduction
 - Convection
 - Active Transport
 - Radiation
- Q.3 How many types of convection process are there? (CO1)
- One
 - Two
 - Three
 - None of the options

Q.4 The process by which a pot of water on your electric stove starts to heat is (CO1)

- a) Conduction b) Convection
- c) Radiation d) Evaporation

Q.5 Which of the following is an example of steady-state heat transfer? (CO1)

- a) Electric bulb cools down by the surrounding atmosphere
- b) Chilling effect of cold wind on a warm body
- c) Boilers and turbines
- d) Cooling of I.C engines

Q.6 The Stefan Boltzman law states that (CO4)

- a) $E \propto T$ b) $E \propto T^2$
- c) $E \propto T^3$ d) $E \propto T^4$

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Driving force of Heat Transfer is _____ (CO1)

Q.8 Solid medium is required for conduction (True/False) (CO2)

Q.9 What is insulator. (CO3)

Q.10 The absorptivity of black body equals to _____ (CO4)

Q.11 For forced convection $\{Nu=f(\text{_____,}Pr)\}$ (CO3)

Q.12 Write the value of Stefan Boltzman constant. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between steady and unsteady state heat transfer? (CO1)

Q.14 Discuss the concept of radiation with example. (CO1)

Q.15 Discuss about the Fourier's law of heat conduction with assumption. (CO2)

Q.16 Write any four physical properties of insulating materials. (CO3)

Q.17 Explain the concept of convection with example. (CO1)

Q.18 Explain the effect of temperature on thermal conductivity of solid, liquid and gas. (CO2)